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2

Pulmonary artery catheter

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Since its inception, the pulmonary artery catheter has enjoyed widespread use in both medical and surgical critically ill patients. It has also endured criticism and skepticism about its benefit in these patient populations. By providing information such as cardiac output, mixed venous oxygen saturation, and intracardiac pressures, the pulmonary artery catheter may improve care of the most complex critically ill patients in the intensive care unit and the operating room. With its ability to transduce pressures through multiple ports, one of the primary clinical uses for pulmonary artery catheters is real-time intracardiac pressure monitoring. Correct interpretation of the waveforms is essential to confirming correct placement of the catheter to ensure accurate data are recorded. Major complications related to catheter placement are infrequent, but misinterpretation of monitored data is not uncommon and has led many to question the utility of the pulmonary artery catheter. The evidence to date suggests that the use of the catheter does not change mortality in many critically ill patients and may expose these patients to a higher rate of complications. However,

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additional clinical trials are needed, particularly in the most complex critically ill patients, who have generally been excluded from many of the research trials performed to date.

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Evolution of the pulmonary artery catheter

Since its inception, the pulmonary artery catheter (PAC) has enjoyed widespread use in both medical and surgical critically ill patients. It has also endured criticism and skepticism about its benefit in these patient populations. The PAC is used to aid in diagnosing and treating the complex nature of cardiopulmonary derangements in patients with critical illness. It also helps to guide the anesthesiologist or intensivist in the treatment of the most critically ill patients who present to the operating room and the intensive care unit (ICU). By providing accurate measurements of a wide range of cardiovascular variables such as cardiac output (CO), mixed venous oxygen saturation, and intracardiac pressures, the PAC may improve care of our most complex and hemodynamically unstable patients.

The idea of right heart catheterization dates back to 1929 when Dr. Werner Forssmann placed a urethral catheter through his own vein to measure pressures in his right heart chambers. The technique progressed further in 1944 when Cournand and Lauson published their recordings of right heart pressures. In 1954, Connolly and colleagues demonstrated that pulmonary artery wedge pressures (PAWPs) correlated closely with left atrial pressures (LAPs), thus further expanding the role of heart catheterization. Forssmann, Cournand, and Richards were awarded the Nobel Prize in Medicine in 1956 for their work with the PAC [1].

Right heart catheterization remained a procedure performed only in the cardiac catheterization laboratory with stiff catheters that could not remain in place for long periods until 1970, when Drs. Swan and Ganz introduced the flow-directed balloon-tipped catheter, which could be placed at the bedside without the use of fluoroscopy. The idea of balloon floatation is said to have come to Dr. Swan while he was watching boats in the Santa Monica Bay, advancing downwind with their spinnakers filled. He believed that by adding a balloon tip, the catheter would be carried into the PA by the continuous blood flow. Dr. Swan and Dr. Ganz published a case series of 100 patients in which they successfully placed the catheter in 60 patients without the use of fluoroscopy [2,3]. They also showed that catheterization was completed far faster than traditional placement without the aid of the balloon-tipped catheter. One of the most significant advantages of the so-called Swan–Ganz catheter was that it could be left in place for days at a time, because the balloon tip was mounted on a flexible catheter, which was theoretically less traumatic than traditional stiffer cardiac catheters. With the balloon deflated, the PAC could remain in place for prolonged periods of patient monitoring without obstructing flow through the PA. The catheter was further modified to enable measurement of all right-sided pressures, PAWP, thermodilution CO, continuous mixed venous oxygen saturation [4], and pacing of the right atrium (RA) and right ventricle (RV).

Clinical use and practical applications

With the ability to transduce pressures through multiple ports, PACs provide intracardiac pressure monitoring at several right heart sites in real time (Fig. 1). When the PAC is in a proper position with its tip in the PA, the central venous pressure (CVP) orifice, which lies 30 cm from the tip of the catheter, records the right atrial pressure (equal to CVP), while the orifice at the distal tip of the catheter measures the pulmonary artery pressure (PAP). One must use caution in assuming the location of the PAC tip resides in the PA, because in the operating room, PACs are frequently inserted partially to a more proximal location for surgical procedures involving the right side of the heart (tricuspid valve, pulmonic valve surgery, or heart transplantation). In these situations, the distal port records CVP rather than PAP.

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The standard insertion of the PAC involves advancing the catheter from a central vein (typically the internal jugular (IJ) or subclavian) until its tip resides in the RA and is recording a standard CVP waveform. Then, the balloon is inflated with 1.5 ml of air, and the catheter is advanced across the tricuspid valve to the RV, and then across the pulmonic valve into the PA (Figs. 1 and 2). It is important to recognize that normal RV and PA pressure values are only slightly different and failure to recognize a PAC tip that is misplaced in the RV will result in diagnostic errors and put the patient at the risk of ventricular arrhythmias and cardiac perforation.

When the PAC, with balloon inflated, is further advanced beyond the proximal PA, it becomes "wedged," thereby isolating the catheter tip from the PA and instead measuring PAWP, also termed pulmonary artery occlusion pressure (PAOP) (Fig. 2).

In addition to the proximal CVP and distal PA ports, some PACs have an additional venous infusion port (VIP) located close to the proximal CVP port. Note that the exact location of the VIP varies depending on the type of PAC. This supplementary venous lumen allows continuous infusions of vasoactive drugs that are commonly administered to patients requiring PAC monitoring, while simultaneously allowing uninterrupted CVP monitoring. In clinical practice, both the CVP and VIP ports are convenient for drug bolus injection and rapid flushing into the central circulation. Other features available in some PACs include external electrodes that allow atrial, ventricular, and atrioventricular sequential endocardial pacing. However, more reliable RV pacing is provided with a dedicated endocardial pacing wire, which can be placed through the PAC introducer sheath after removal of the PAC.

The utility of PACs goes beyond that of continuous intracardiac pressure monitoring, as one of the most common indications for placement of a PAC is the measurement of CO. This is typically measured by cold bolus thermodilution or continuous warm thermodilution [5,6]. In addition, some PACs can continually measure PA blood oxygen saturation, or mixed venous saturation (SvO_2) using reflectance oximetry measured with a tiny light source near the catheter tip. Frequently, patients requiring PACs in

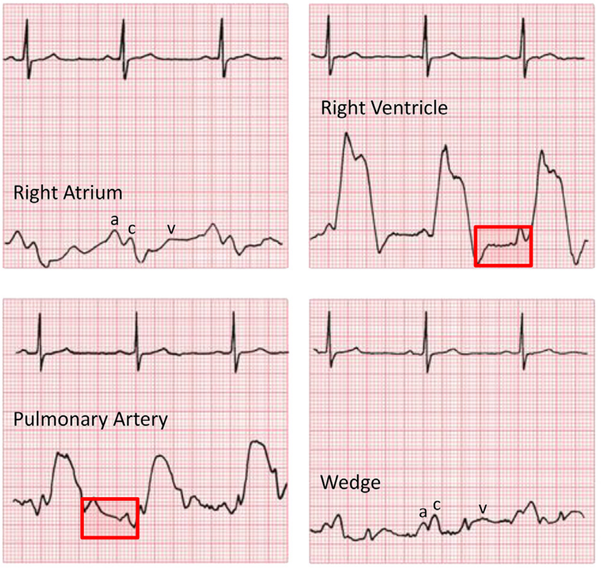

Fig. 1. Characteristic pressure waveforms of each cardiac chamber seen during advancement of a pulmonary artery catheter. The CVP/RA wave demonstrates three peaks (*a*, *c*, and *v* waves) and two descents (*x* and *y*). As a significant increase in systolic pressure is seen in the RV as compared to the CVP, sequentially, the PA waveform shows a diastolic "step-up." Note the diastolic waveform contour can assist in differentiating RV from PA pressures. There is a gradual increase in RV pressures during diastole, while the PA shows a continued decrease in pressures throughout diastole. Finally, advancement into the "wedge" position yields a waveform similar to that of CVP, which is delayed (compared to the EKG) secondary to the transit time through the pulmonary vasculature.

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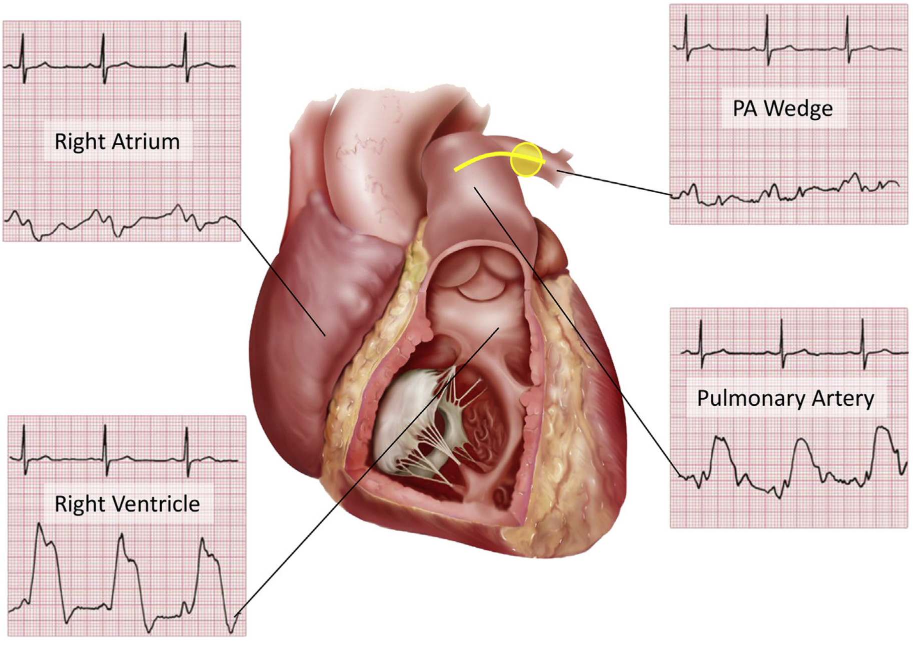

Fig. 2. Representation of each pressure waveform as the PA catheter moves through the right-sided cardiac chambers to its final location in the pulmonary artery.

the operating room or ICU require continuous monitoring of one or perhaps all the above parameters to make appropriate clinical decisions and interventions.

Insertion of PACs

Prior to placement of the PAC, sterile cannulation of the IJ vein or subclavian vein must be performed in order to place an introducer sheath. Although it is possible to cannulate more distal veins (i.e., femoral or brachial) for insertion of the catheter, this increases the level of difficulty for flotation of the catheter tip into the PA owing to the greater distance that the catheter must travel in the venous system as well as a more acute angle to enter the RA from the femoral approach. Venous access through

Fig. 3. Surface ultrasound images demonstrating needle puncture of the internal jugular vein and subsequent wire placement into the vessel. The wire can be seen in the vein adjacent (anterior/lateral) to the internal carotid artery in a short axis. A long-axis view of the wire in the internal jugular is then obtained to confirm travel within the vessel lumen prior to dilation and final placement of a pulmonary artery catheter introducer.

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the right IJ vein provides the most direct route for flow-directed placement of a PAC and is therefore considered the first choice by most clinicians. Recently published American Society of Anesthesiologists (ASA) practice guidelines for central venous cannulation recommend ultrasound guidance for all elective IJ vein catheterizations (Fig. 3) (Level A1 evidence) [7].

Once the desired vein is cannulated, the PAC is placed through a sterile sheath and each of the ports is flushed with saline to confirm patency and expel air. Next, the balloon at the distal tip of the catheter is inflated with 1.5 cc of air to check that it is intact and symmetrically extending just beyond the catheter tip, and then it is deflated fully. Once the catheter tip is inserted beyond the end of the introducer sheath (typically at a depth of 15–20 cm, depending on the length of the introducer sheath), the balloon is inflated, noting any increased resistance compared to the "test" inflation. Resistance may indicate the balloon is still within the introducer sheath or that the PAC tip is otherwise improperly located. Once the balloon is successfully inflated, normal blood flow through the right heart chambers will guide the tip of the catheter into the PA. Note that the PAC should only be advanced with the balloon inflated to minimize the risk of myocardial or valvular injury from the unprotected catheter tip.

As the PAC passes through the RA, RV, and into the PA, each chamber has defining hemodynamic characteristics as seen on a pressure waveform, which should be used to "float" the PAC (Figs. 1 and 2). The normal hemodynamic values for these variables are summarized in Table 1. Fluoroscopy or transesophageal echocardiography guidance can complement pressure monitoring during PAC placement in the event of difficulty. Depending on the location chosen for central venous cannulation, each of the abovementioned waveforms should be encountered sequentially at a typical distance from the cannulation site. Failure to achieve this indicates catheter misdirection (most commonly down the inferior vena cava rather than across the tricuspid valve, particularly when the PAC is inserted from the right IJ vein), or coiling within the RV, which can lead to catheter knotting if not recognized. If the expected pressure tracings are not recorded at the typical depth of insertion, the PAC balloon should be deflated and the catheter withdrawn into the RA to attempt repeat flotation. Placing the patient in the Trendelenburg position with the left side up can facilitate catheter passage across the tricuspid valve, and the head-up position often makes crossing the pulmonic valve easier. From the right IJ vein, RA pressure or CVP should be recorded at an insertion depth of 20–30 cm, the RV at 30–40 cm, the PA at 40–50 cm, and a wedge pressure at distances slightly beyond 50 cm.

The use of PACs is not without complications. These complications stem from three sources: venous cannulation, PAC placement or manipulation, and errors in monitoring. Central venous catheter placement complications include bleeding, pneumothorax, hemothorax, nerve injury, infection, venous thromboembolism, and venous air embolism. Those that arise from PACs specifically include arrhythmias, mechanical damage to the cardiac valves, RA, RV or PA, thrombosis, and infection. During flotation of a PAC, transient arrhythmias, including premature ventricular contractions, are common and occur in up to 70% of patients. Pulmonary infarction or PA rupture may result from a catheter that continuously remains in the wedged position or a catheter that is inserted into a small branch PA. A chest X-ray is always obtained following PAC placement, and the catheter tip should be identified no more than a 3–5 cm from the midline or within 2 cm of the cardiac border [8]. PA rupture is the most serious life-threatening complication of catheter use, carrying a 70% mortality. Some authors have suggested that pulmonary hypertension and patients receiving anticoagulants or who are otherwise

Table 1
Normal intracardiac pressures and pulmonary artery catheter derived values.

Parameter	Range (Mean)
Central venous/right atrial pressure	1–5 (3) mmHg
Right ventricular systolic pressure	15–30 (25) mmHg
Right ventricular diastolic pressure	1–7 (6) mmHg
Pulmonary artery systolic pressure	15–30 (25) mmHg
Pulmonary artery diastolic pressure	4–12 (9) mmHg
Pulmonary artery mean pressure	9–19 (15) mmHg
Pulmonary artery wedge pressure	4–12 (9) mmHg
Mixed venous oxygen saturation	65–75%
Cardiac output: cardiac index	4–8 L/min: 2.5–4 L/min/m^2

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coagulopathic are at a higher risk of rupture [9]. Other complications of PAC use include data misinterpretation and the inappropriate therapeutic interventions resulting from these errors. Careful attention to the details of PAC placement and management, including proper pressure transducer calibration and leveling, will reduce complications.

Interpretation of PAC data

As the PAC is advanced through the RA, RV, and into the PA, each distinct pressure waveform must be recognized and interpreted correctly. (Figs. 1 and 2) The initial pressure tracing seen will be that of the CVP, a waveform that consists of three peaks (*a*, *c*, and *v*) and two descents (*x* and *y*) (Fig. 4). Note that the CVP and RA pressure waveforms are considered interchangeable, both quantitatively and

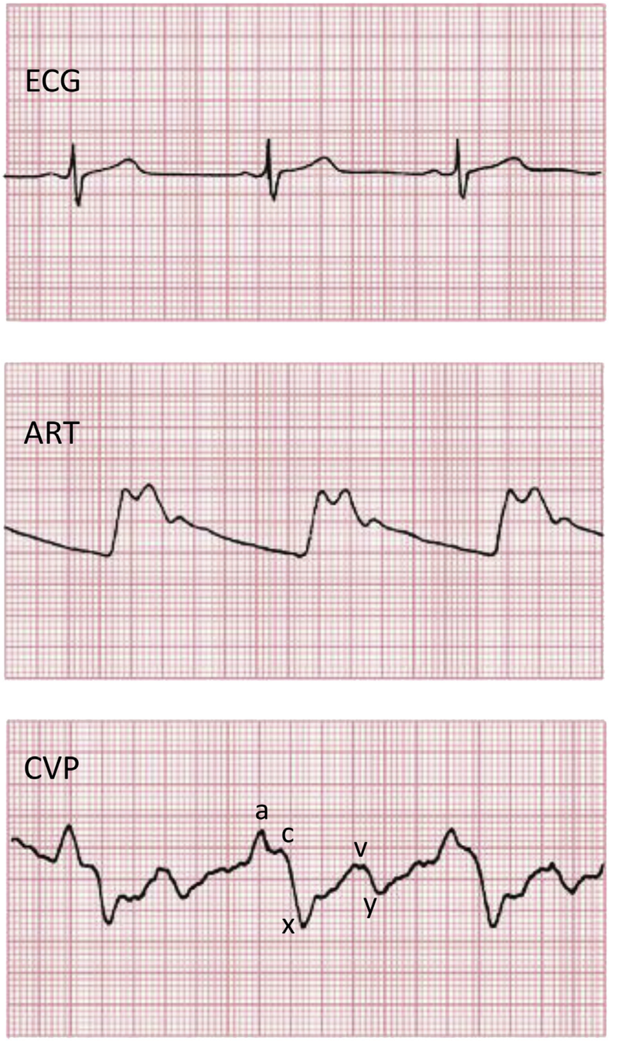

Fig. 4. Normal CVP tracing correlated with arterial blood pressure and EKG. Wave identification is significantly aided by the timing in relationship to the EKG P wave and QRS complex which denote the electrical stimulus. The CVP a wave will follow atrial systole (EKG P Wave), and the CVP c wave will follow the QRS complex in early systole.

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qualitatively, with the former being recorded in the superior vena cava and the latter recorded in the RA. The *a wave* results from atrial contraction causing an increase in RA pressure during the late filling portion of diastole. This is seen after electrical depolarization of the atria as noted on the electrocardiogram (ECG) as the P wave. Following end diastole, the atrium relaxes causing a decline in atrial pressure, which is interrupted by the second peak, or *c wave*. Isovolumetric RV contraction causes tricuspid annular movement away from the RV apex towards the RA. The *c wave* follows the onset of the ECG QRS complex denoting early systole. Atrial pressures continue to decline during mid-systole, seen as the *x descent*. This can be further divided into x and x', the later comprising the portion of atrial relaxation occurring after the *c wave*. The final pressure peak seen on the CVP tracing is the *v wave*. Venous filling of the RA through the superior and inferior vena cavae produces the *v* wave during mid- to late systole. The ECG T wave occurs prior to the peak of the *v wave*. Finally, RA pressure decreases with the *y descent* as the tricuspid valve opens and blood flows down its pressure gradient into the RV. This is the early portion of ventricular diastole and accomplishes the majority of diastolic filling prior to the "atrial kick" or *a* wave.

When the PAC is advanced from the RA, it should cross the tricuspid valve and record an RV pressure waveform. Ventricular arrhythmias are common when the PAC tip is in the RV and are typically benign and self-limited. However, if there is sustained ventricular ectopy, prompt advancement of the catheter tip into the PA or balloon deflation and catheter withdrawal into the RA may be necessary. An increase in systolic pressure indicates the catheter's arrival in the RV (Figs. 1 and 2). In addition, during diastole, the pressure tracing will have an up-sloping contour resulting from diastolic RV filling. The shape of the RV pressure waveform is in distinct contrast to the PA pressure waveform, which has a down-sloping diastolic contour.

From the RV, the PAC is advanced across the pulmonic valve into the main PA. This is seen as an increase or "step-up" in the diastolic pressure in the PA compared to the RV, and a dicrotic notch in the PA pressure trace, which occurs with pulmonic valve closure.

Finally, the PAC is advanced further until the balloon occludes a branch of a central PA and produces a characteristic "wedge" or PAWP tracing. Balloon occlusion of the branch PA isolates the catheter tip from upstream PA pressure, and this creates a static "column" of blood connecting the wedged catheter tip to the left atrium (LA), by way of the interposed segment of the pulmonary vasculature (Fig. 5). Thus, the PAWP is an indirect measurement of LAP, and normal wedge pressure waveforms contain the same characteristic wave components as CVP or RAP pressures. In contrast to RA pressure, which is directly recorded by the PAC lumen in the RA, LAP is measured indirectly by the wedged PAC, resulting in a 150–200-ms delay owing to retrograde transmission of the LAP waveform through the pulmonary vascular bed. As a result, the PAWP *a wave* appears to follow the ECG R wave, rather than immediately following the ECG P wave, like the *a wave* in the CVP waveform. Mean rather than phasic PAWP values

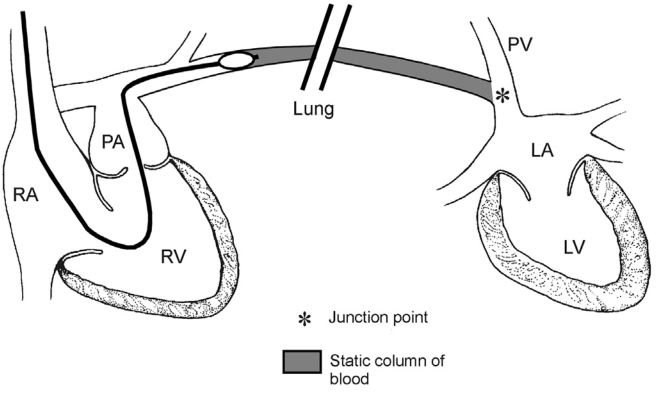

Fig. 5. Illustration of PAC balloon in the "wedged" position in a branch of the main pulmonary artery. In this position, the transducer at the tip of the catheter is isolated from forward pulmonary artery blood flow. The reflected waveform measured is a delayed representation of pulmonary venous and left atrial pressures.

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are displayed numerically on the bedside monitor screen. Although these values are typically used as an approximation for left ventricular (LV) end-diastolic (or LV filling) pressure, the most accurate estimate for LV end-diastolic pressure is the peak of the PAWP *a wave*, which is inscribed at end diastole, just prior to mitral closure and onset of LV systolic contraction.

Analysis of the aforementioned waveforms may reveal abnormalities, which provide diagnostic clues about hemodynamic abnormalities, including arrhythmias and valvular dysfunction. Atrial fibrillation causes a loss of the CVP (and PAWP) *a wave* and a prominent *c wave*. The absence of a coordinated atrial contraction causes increases in RA volume (and pressure) at end diastole making for an exaggerated *c wave* (Fig. 6). Atrioventricular nodal or junctional rhythms may cause the RA to contract against a closed tricuspid valve during ventricular systole resulting in large or cannon *a waves* (Fig. 7). Cannon waves may be seen whenever atrioventricular dissociation is present, such as in patients with heart block. CVP tracings with a tall systolic *c*–*v wave* are characteristic of significant tricuspid regurgitation (Fig. 8). This wave typically obliterates the systolic *x descent* as a consequence of RV regurgitant flow into the lower-pressure RA during systole. Because the mean RA pressure or CVP is increased by this regurgitant *c*–*v* wave, the true end-diastolic RV-filling pressure is overestimated by the mean CVP. Instead of relying on the displayed mean CVP value, RV end-diastolic pressure is best measured by the phasic pressure recorded at end diastole, at the time of the ECG R wave. Similarly, a tall systolic *c*–*v wave* recorded in a PAWP tracing is the hallmark of severe mitral regurgitation.

Variations in intracardiac pressure tracings are noted during the respiratory cycle and caused by changes in intrathoracic pressures. Because of these variations, end-expiratory pressures should be used to provide the best estimates of transmural filling pressures. During both spontaneous and controlled mechanical ventilation, end-expiratory intrathoracic pressures most closely approximate atmospheric pressure, thereby minimizing changes in transmural pulmonary and pericardial pressures and their influences on the recorded intravascular pressure, CVP or PAWP [10].

As a continuous pressure-monitoring device, the PAC provides invaluable information regarding temporal trends in intracardiac and central vascular pressures. One particular example is the utility of the PAC to diagnosis RV failure. By following simultaneous trends of CVP and mean PA pressure, a clinician can identify incipient RV dysfunction by noting a rising CVP in the setting of normal or falling PA pressure. By contrast, an increase in both CVP and PA pressure more often indicates LV dysfunction, increased pulmonary vascular resistance, or an increase in circulating intravascular volume. The ability

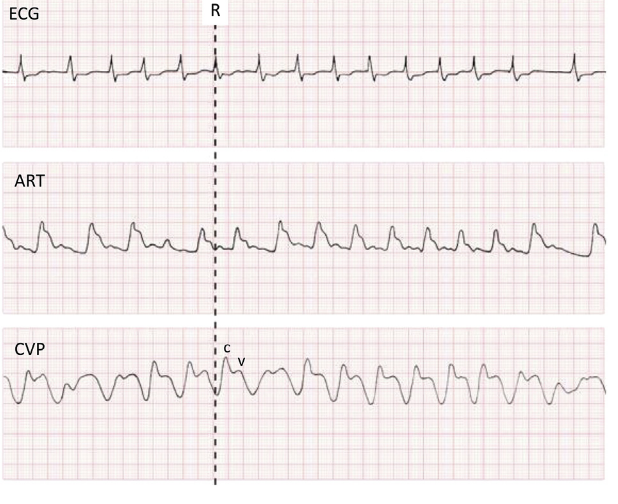

Fig. 6. Atrial fibrillation and corresponding EKG, arterial waveform, and CVP waveform. Note the characteristic loss of P wave on EKG, and varying QRS durations. As a result, there is a loss of coordinated atrial contraction as represented by a loss of the a wave on the CVP tracing.

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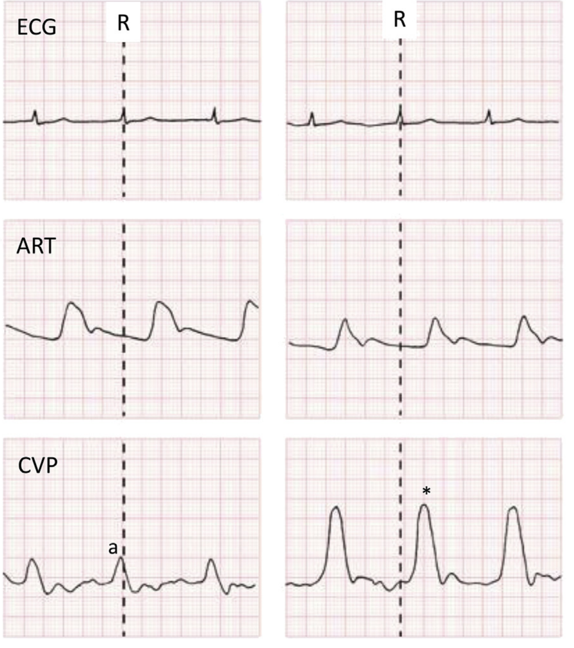

Fig. 7. AV dissociation (including junctional rhythms) results in cannon a waves as seen on this CVP tracing. Atrial contraction against a close tricuspid valve during systole results in a greater rise in atrial pressure and retrograde blood flow as represented here.

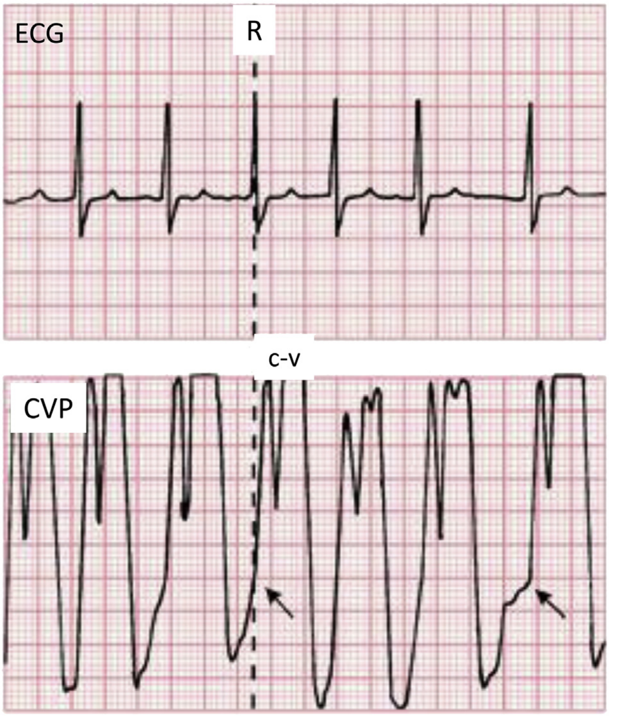

Fig. 8. Tricuspid regurgitation seen on CVP tracing, resulting in a large $c-v$ wave occurring during systole.

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of the PA catheter to monitor pressures proximal and distal to the RV makes it uniquely suited to distinguish these situations from one another.

The utility of PACs extends beyond pressure monitoring and includes the ability to measure CO and SvO_2. PACs use thermodilution techniques to measure right-sided CO. Intermittent CO measurements are obtained by injecting room-temperature saline or dextrose as a bolus through the proximal RA port of the catheter. Mixing of the cooler intravenous (IV) injection with blood from the RA and RV slightly decreases the blood temperature recorded as it passes beyond a thermistor near the tip of the catheter in the PA. A temperature versus time washout curve is thus created, as the blood flows past the PA thermistor. This is used to calculate the temperature change over time, which is inversely proportional to RV output. Assuming no intracardiac shunting, the RV output equals the systemic CO. Significant tricuspid regurgitation and intracardiac shunts will give misleading CO values using this technique, the former caused by reflux of cold injectate back into the RA, and the latter by producing a physiologic difference between RV and LV outputs. Continuous cardiac output (CCO) catheters use a heating filament affixed to the catheter approximately 20 cm from its tip to warm blood in the RA and RV as it flows across the tricuspid valve. These tiny temperature changes are sensed by the thermistor near the tip of the PAC, and using a sophisticated algorithm, CO is calculated by "warm" thermodilution washout. Although this information is updated every 30–60 s, the CO displayed is a running average of values collected over a several-minute interval and consequently will not indicate abrupt changes in LV output. As bolus CO and CCO measurements are both based on a thermal signal, similar issues can confound both types of CO monitoring, including peripheral fluid boluses and other thermal signal errors [11,12].

Some PACs are equipped with fiber-optic monitoring of SvO_2 by recording a reflectance oximetry signal from PA blood. True SvO_2 should be recorded from the PA rather than from a central vein or the RA, because the latter do not incorporate returning coronary venous blood, which has a low saturation of ~25% due to high myocardial extraction of oxygen. Typically, the "oximetric" PAC is calibrated in vitro prior to insertion, but it can also be recalibrated in vivo by drawing a PA mixed venous blood sample to use as the reference value. Care must be taken to avoid sampling blood for calibration when the PAC is in the wedged position, because blood drawn from the wedged catheter tip will be "arterialized" pulmonary capillary blood rather than mixed venous PA blood. Because fiber-optic PA SvO_2 monitoring is continuous rather than intermittent or averaged over time, changes in the saturation generally precede hemodynamic changes and may therefore be a useful early warning sign of hemodynamic deterioration. Although changes in SvO_2 are often considered to be directly related to changes in CO, the relationship between these variables is more complex. The Fick equation describes the relationship between oxygen consumption, CO, and the arteriovenous oxygen content difference. Rearranging the equation shows that $SvO_2 = SaO_2 - (VO_2/[CO \times 1.36 \times Hgb])$, where SaO_2 is the arterial hemoglobin saturation (%), VO_2 is the oxygen consumption (ml/min), CO is cardiac output (ml/min), and Hgb is hemoglobin concentration (g/dl). Based on this relationship, it is clear that a decrease in SvO_2 can result from a decrease in CO, but also be caused by a decrease in hemoglobin, a decrease in arterial hemoglobin saturation, or an increase in oxygen consumption.

Evidence regarding pulmonary catheter use

The flow-directed PAC became prevalent in the ICU because of its quick and relatively easy insertion as compared to standard cardiac catheterization techniques used in the cardiology catheterization suite. Furthermore, the wealth of hemodynamic variables available to supplement standard clinical measures and guide diagnostic decision making seemed intuitively appealing and physiologically sound. The use of the catheter became widespread in both cardiac and noncardiac critically ill patients, but the rapid growth and widespread use were controversial from the outset. Many early retrospective studies revealed excess mortality in patients receiving PAC monitoring, but these studies were criticized for their design. For example, in 1996, a highly publicized nonrandomized trial, SUPPORT, was conducted in five major medical centers and careful propensity matched patients monitored with PACs to other ICU patients. The comparison showed that the use of the PAC was associated with increased mortality, increased hospital length of stay, and increased hospital costs [13]. However, PAC use remained widespread because randomized prospective clinical trial data were not yet available.

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Over the next several years, several prospective randomized trials were completed and revealed no difference in mortality in critically ill patients monitored with PACs as compared to patients without PAC monitoring. While the excess mortality reported in the Surfactant Positive Airway Pressure and Pulse Oximetry Trial (SUPPORT) was not substantiated, most of these randomized trials showed an increase in complications related to PAC monitoring. For example, in 2003, a randomized controlled trial looking at goal-directed therapy guided by PAC monitoring in ASA III and IV patients undergoing urgent or elective surgery revealed identical survival between PAC-monitored patients and controls. This study did not find increased mortality in the PAC-monitored group, but it did reveal more pulmonary complications in these patients. Of note, centers in this study only enrolled a mean of one patient per month and management between the two groups was similar regardless of the data collected [14]. Another large randomized prospective study failed to show a difference in hospital mortality in patients undergoing PAC monitoring compared with control patients; however, this trial also revealed a higher incidence of complications in the PAC-monitoring group, although no specific protocol was used to help guide clinicians in interpreting the PAC data. This study, known as the PAC-Man study, enrolled 1041 patients across 65 ICUs in the UK [15]. The Evaluation Study of Congestive Heart Failure and Pulmonary Artery Catheterization Effectiveness (ESCAPE) trial is a third randomized prospective study of critically ill patients, but this trial focused on patients with severe congestive heart failure. Once again, the investigators found no effect on mortality between PAC-monitored patients and controls [16].

Of note, however, at the same time the ESCAPE randomized trial was performed, a registry was also collected including patients who were not randomized and entered into the trial. Interestingly, these registry patients were sicker, on average, than those studied in the trial, and their mortality was greater than that seen in trial patients. It is quite possible that registry patients monitored with PACs would have suffered even higher mortality had they been monitored without this catheter [16,17]. A Cochrane review of PAC monitoring reported in 2013 identified 13 relevant trials and included a total of 5686 patients. The review concluded that the use of PAC monitoring did not alter the mortality, length of stay, or cost for adult intensive care patients. However, the review highlights the fact that available studies have not focused on the highest risk and the most severely ill patients (including complex cardiac surgical patients), and suggests that additional clinical trials are needed to assess PAC-guided management protocols in specific high-risk patients. These protocols could result in benefits such as reversal of shock, improvement in organ function, and reduction in vasopressor use [18].

While these studies have not demonstrated a difference in mortality when patients are monitored with PACs, trials to date have excluded patients who might show the greatest benefit from PA catheterization. In fact, the most recent American College of Cardiology/American Heart Association practice guidelines for coronary artery bypass graft surgery continue to recommend PAC use in patients presenting with cardiogenic shock (Class I, Level C Evidence) and patients with acute hemodynamic instability (Class IIA, Level B Evidence) [19]. An examination of the Acute Decompensated Heart Failure Syndromes (ATTEND) Registry, which focuses on patients with severe heart failure, found that the appropriate use of a PAC effectively decreased in-hospital mortality in patients with acute heart failure symptoms, particularly in those with lower systolic blood pressure or receiving inotropic therapy. These data suggest that PACs may have benefit in specific patient populations and further study is warranted [20].

Given that the available clinical trials, particularly the few randomized prospective trials, provide no evidence for the benefit of monitoring patients with PACs, it is not surprising that the use of catheters has decreased considerably over the past 10–20 years. Data derived from the National Inpatient Sample between 1993 and 2004 showed a 65% decrease in the use of PACs in hospitalized medical and surgical patients [21]. A retrospective cohort study of >300,000 ICU patients cared for in hundreds of ICUs from 2001 to 2008 revealed that PAC use decreased from 11% of ICU patients to 6% of patients. Further multivariate regression analysis found that surgical patients and those in surgical ICUs continue to have a higher rate of PAC use [22].

Today, PACs are used less commonly than in the past two decades, and it is likely that this trend will continue. However, intensive care physicians, anesthesiologists, and surgeons who care for the most critically ill surgical patients, particularly those undergoing complex cardiac operations, still employ PAC monitoring regularly. Until additional trials in these patient groups are conducted, or alternative monitoring techniques are available, the PAC appears to still have a role in clinical care.

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Practice points

- Pulmonary artery catheters should be considered in the most complex critically ill cardiac and noncardiac patients when variables such as cardiac output, mixed venous oxygen saturation, and "wedge" pressure may redirect clinical decision making in the intensive care unit and operating room.
- Pulmonary artery catheters should be inserted using ultrasound guidance when available and care must be taken to maintain strict sterility during placement.
- Practitioners should be familiar with the pressure waveforms associated with pulmonary artery catheters to ensure proper placement and interpretation of the data.

Research agenda

- We need randomized controlled trials in the most complex cardiac and noncardiac patients to determine the true utility of pulmonary artery catheters.
- We need to determine functional protocol-guided use of pulmonary artery catheters to standardize use in the most at-risk patients.

Conflict of interest statement

None.

References

[1] Chatterjee K. The Swan-Ganz catheters: past, present and future. A viewpoint. Circulation 2009;119:147–52.
[2] Bayliss M, Andrade J, Heydari B, et al. Jeremy Swan and the pulmonary artery catheter: paving the way for the effective hemodynamic monitoring. BC Medical Journal 2009;51:302–7.
[3] Swan HJ, Ganz W, Forrester J, et al. Catheterization of the heart in man with use of a flow-directed balloon-tipped catheter. N Engl J Med 1970;283:447–51.
[4] Ganz W, Donoso R, Marcus HS, et al. A new technique for measurement of cardiac output by thermodilution in man. Am J Cardiol 1971;4:392–6.
*[5] Reuter DA, Huang C, Edrich T, et al. Cardiac output monitoring using indicator-dilution techniques: basics, limits and perspectives. Anesth Analg 2010;110:799–811.
[6] Yelderman M. Continuous measurement of cardiac output with the use of stochastic identification techniques. J Clin Monit 1990;61:322–32.
*[7] Rupp SM, Apfelbaum JL, Blitt C, et al. Practice guidelines for central venous access: a report by the American Society of Anesthesiologists Task Force on Central Venous Access. Anesthesiology 2012;116:539–73.
[8] Houghton D, Cohn S, Schell V, et al. Routine daily chest radiography in patients with pulmonary artery catheters. Am J Crit Care 2002;11:261–5.
[9] Bussieres JS. Iatrogenic pulmonary artery rupture. Curr Opin Anaesthesiol 2007;20:48–52.
*[10] Teplick RS. Measuring central vascular pressures: a surprisingly complex problem. Anesthesiology 1987;67:289–91.
[11] Siegel LC, Hennessy MM, Pearl RG. Delayed time response of the continuous cardiac output pulmonary artery catheter. Anesth Analg 1996;83:1173–7.
[12] Gardner RM. Continuous cardiac output: how accurately and how timely? Crit Care Med 1998;26:1302–3.
[13] Connors AF, Speroff T, Dawson NV, et al. The effectiveness of right heart catheterization in the initial care of critically ill patients. JAMA 1996;276:889–97.
*[14] Sandham JD, Douglas R, Hull D, et al. A randomized, controlled trial of the use of pulmonary-artery catheters in high-risk surgical patients. N Engl J Med 2003;12:5–14.
*[15] Harvey S, Harrison DA, Singer M, et al. Assessment of the clinical effectiveness of pulmonary artery catheters in management of patients in intensive care (PAC-Man): a randomized controlled trial. Lancet 2005;366:472–7.
*[16] Binanay C, Califf RM, Hasselblad V, et al. Evaluation study of congestive heart failure and pulmonary artery catheterization effectiveness: the ESCAPE trial. JAMA 2005;294:1625–33.
*[17] Allen LA, Rogers JG, Warnica JW, et al. high mortality without ESCAPE: the registry of heart failure patients receiving pulmonary artery catheters without randomization. J Cardiac Fail 2008;14:661–9.

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*[18] Rajaram SS, Desai NK, Kalra A, et al. Pulmonary artery catheters for adult patients in intensive care. Cochrane Database Syst Rev 2013;2:CD003408. http://dx.doi.org/10.1002/14651858.CD003408.pub3.
*[19] Hillis LDL, Smith PKP, Anderson JLJ, et al. Special Articles: 2011 ACCF/AHA guideline for coronary artery bypass graft surgery: executive summary: a report of the American College of Cardiology Foundation/American Heart Association task force on practice guidelines. Anesth Analg 2012;114:11–45.
*[20] Sotomi Y, Sato N, Kajiimoto K, et al. Impact of pulmonary artery catheter on outcome in patients with acute heart failure syndromes with hypotension or receiving inotropes: from the ATTEND Registry. Int J Cardiol 2014;172:165–72.
[21] Berthiaume LR, Peets AD, Schmidt U, et al. Time series analysis of use patterns for common invasive technologies in critically ill patients. J Crit Care 2009;24. -0.
[22] Gershengorn HB, Wunsch H. Understanding changes in established practice: pulmonary artery catheter use in critically ill patients. Crit Care Med 2013;41:2667–76.

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